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Gabriele Steidl
Jannis Chemseddine
TU Berlin

Lecture 1: Optimal Transport

Lecture 2: Generative Flows

Lecture 3: From 1D Processes to Flows

Lecture 4: Bayesian Inverse Problems

3. Bayesian Inverse Problems

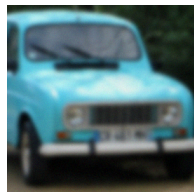
Inverse Problem for a **certain class of images**:

$$y = \text{noisy}(F(x))$$

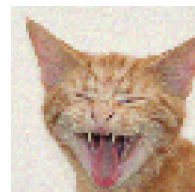
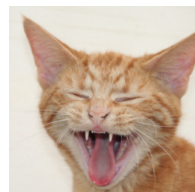
with **ill-posed forward operator** $F : \mathbb{R}^d \rightarrow \mathbb{R}^n$

Typical examples with linear operator $F(x) = Ax$:

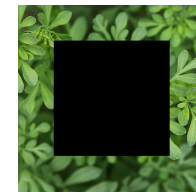
- ◆ Computerized Tomography (CT)
- ◆ Magnetic Resonance Imaging (MRI)
- ◆ image restoration (denoising, deblurring, inpainting, superresolution)



Deblurring



Superresolution



Inpainting

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3. Bayesian Inverse Problems

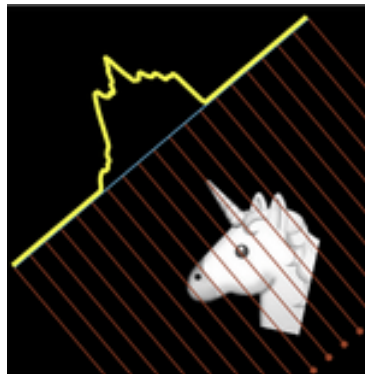
Problem: Well-posedness/conditioned of problem:

- ◆ has solution
- ◆ solution is unique
- ◆ solution depends continuously/stable on input y

Example: Computer Tomography



- ◆ Computation of **sinograms** of Einhorn



History:

- ◆ 1917 J. Radon: from sinogram to image (Radon transform)
- ◆ 1979 G. Hounsfield and A. Cormack: Nobel Prize in (Computertomographie)

Variational Problem: find minimizer of

$$\mathcal{J}(x) = \underbrace{\mathcal{D}(F(x), y)}_{\text{data term}} + \lambda \underbrace{\mathcal{R}(x)}_{\text{regularizer, prior}}, \quad \lambda > 0$$

- ◆ data term $\mathcal{D}$ depends usually on the noise model via $-\log p_{Y|X=x}$
- ◆ prior includes knowledge on the distribution of p_X

Additive Gaussian noise:

$$y = Ax + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2 I_d)$$

$$\mathcal{D}(F(x), y) = \frac{1}{2} \|Ax - y\|^2$$

Bayesian Inverse Problems: sample from the posterior $P_{X|Y=y}$, where

$$Y = F(X) + \Xi$$

Bayesian law: $p_{X|Y=y}(x) = p_{Y|X=x}p_X(x)/p_Y(y)$

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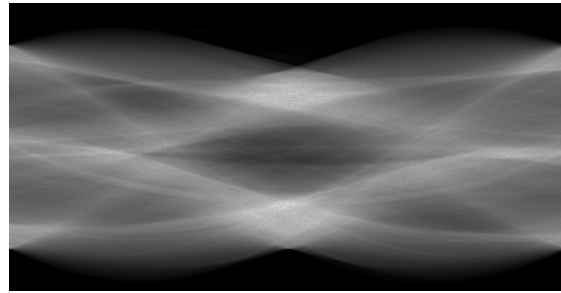
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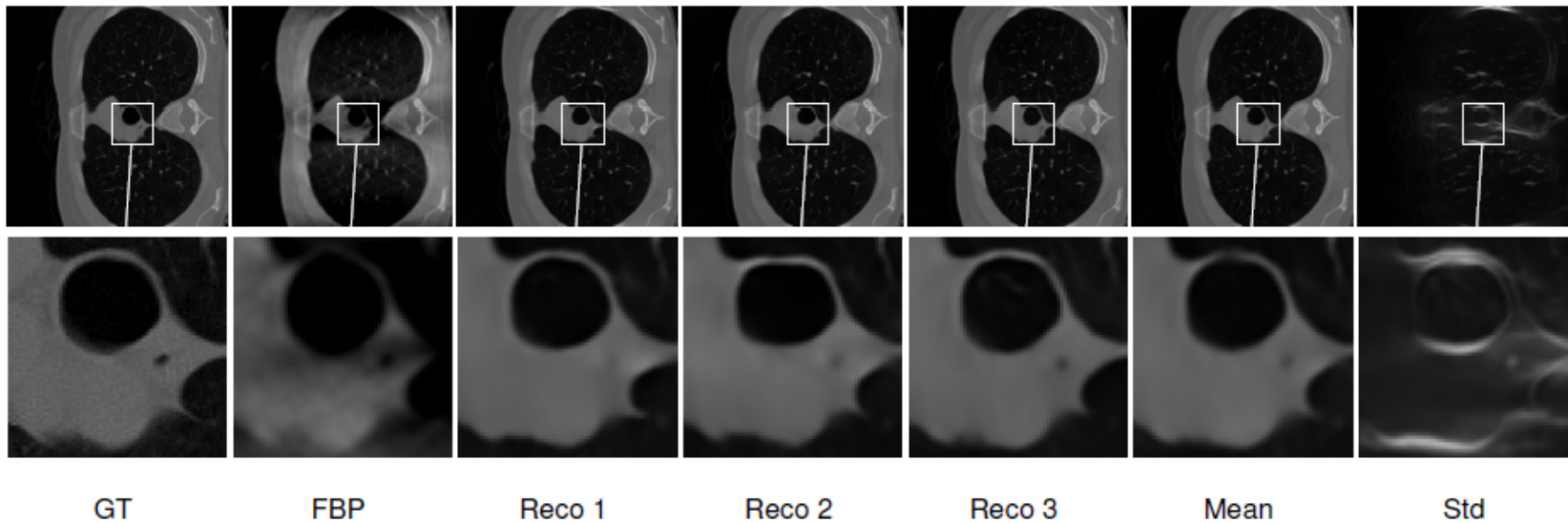
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„Limited Angle” CT



Sinogramm

Bedingte Bildgenerierung: Unsicherheiten in Bildgenerierung werden sichtbar



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3. Bayesian Inverse Problems

1. Bayesian OT Flow Matching

2. PnP Flow

Ref: J. Chemseddine, P. Hagemann, G. Steidl, C. Wald,
Conditional Wasserstein distances with applications in Bayesian OT flow
matching,
Journal of Machine Learning Research (JMLR) 26 (141), 1-47

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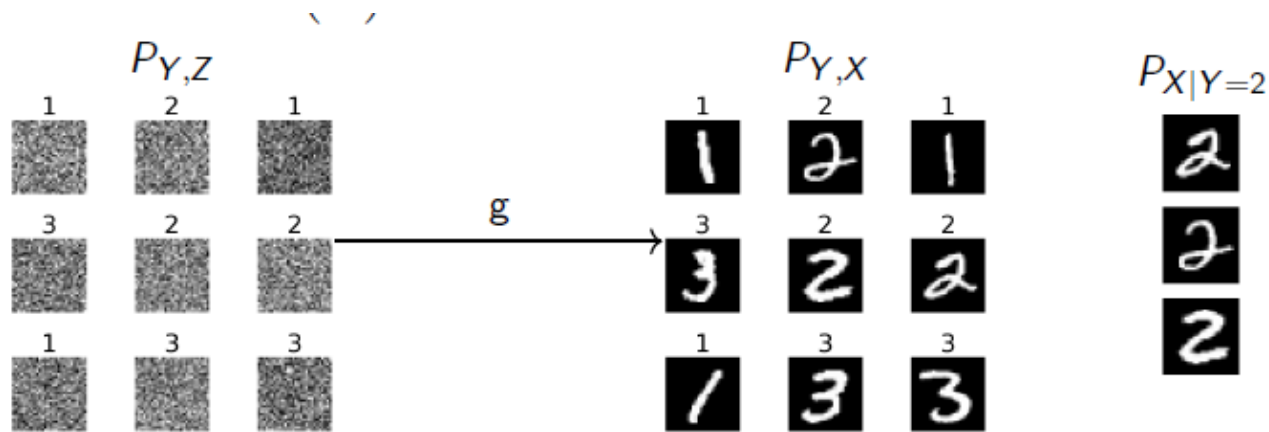
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Conditional Wasserstein Distances



We learn $g^\theta : [0, \dots, 9] \times \mathbb{R}^d \rightarrow [0, \dots, 9] \times \mathbb{R}^d$ such that

$$g_\#^\theta P_{Y,Z} = P_{Y,X}$$

but we want to sample from the posterior $P_{X|Y=y}$.

"Chain rule" of Kullback-Leibler Divergenz (KL):

$$\text{KL}(P_{Y,X}, P_{Y,Z}) = \mathbb{E}_{y \sim P_Y} [\text{KL}(P_{X|Y=y}, P_{Z|Y=y})]$$

does not hold true for Wasserstein distances !

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Conditional Wasserstein distance

$$W_{p,Y}(P_{Y,X}, P_{Y,Z}) := \left(\inf_{\alpha \in \Gamma_Y^4} \int_{(A \times B)^2} \|(y_1, x_1) - (y_2, x_2)\|^p d\alpha \right)^{\frac{1}{p}}$$

with Y -diagonal couplings $\Gamma_Y^4 := \left\{ \alpha \in \Gamma(P_{Y,X}, P_{Y,Z}) : \pi_{\#}^{1,3} \alpha = \Delta_{\#} P_Y \right\}$ on the **conditional Wasserstein space**

$$\mathcal{P}_{p,Y}(A \times B) = \{ \mu \in \mathcal{P}_p(A \times B) : \pi_{\#}^A \mu = P_Y \}$$

Then we have the "chain rule":

$$W_{p,Y}^p(P_{Y,X}, P_{Y,Z}) = \mathbb{E}_{y \sim P_Y} [W_p^p(P_{X|Y=y}, P_{Z|Y=y})]$$

Curves in conditional Wasserstein space $\mathcal{P}_{p,Y}$:

- ◆ $e_t(\mathbf{y}_1, x_1, \mathbf{y}_2, x_2) := (1 - t)(\mathbf{y}_1, x_1) + t(\mathbf{y}_2, x_2)$
- ◆ $\mu_t = (e_t)_{\#} \alpha$, $\alpha \in \Gamma_Y^4(\mu_0, \mu_1)$ is a **curve in $W_{2,Y}$**
- ◆ $L(\theta) = \mathbb{E}_{t \sim [0,1], (\mathbf{y}_1, x_1, \mathbf{y}_2, x_2) \sim \alpha} [\|u(t, e_t(\mathbf{y}_1, x_1, \mathbf{y}_2, x_2), \theta) - (0, x_2 - x_1)\|^2]$

Drawback of $W_{p,Y}$: not implemented in standard OT libraries

Y - penalized Wasserstein distance

$$W_{p,\beta}(\mu_0, \mu_1)^p := \inf_{\alpha \in \Gamma(\mu_0, \mu_1)} \int_{(A \times B)^2} d_\beta^p((y_1, x_1), (y_2, x_2)) \, d\alpha$$

$$d_\beta^p((y_1, x_1), (y_2, x_2)) := \|x_1 - x_2\|^p + \beta \|y_1 - y_2\|^p$$

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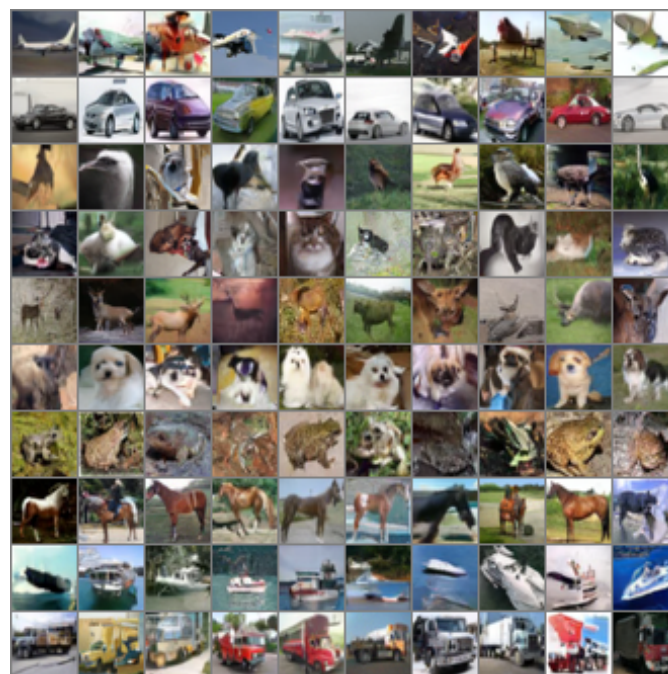
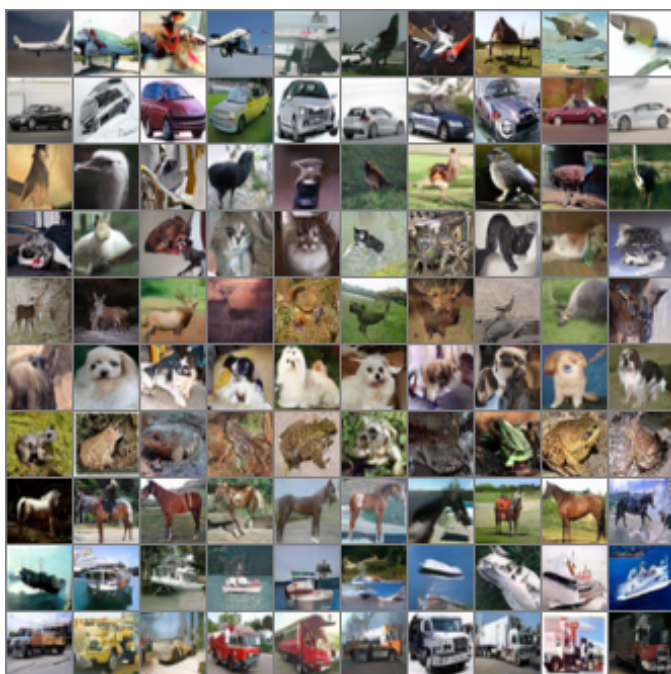
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Bayesian OT Flow Matching: Conditional Wasserstein Distances



CIFAR10, $\beta = 1, 3, 20$ and diagonal, $d = 3072$

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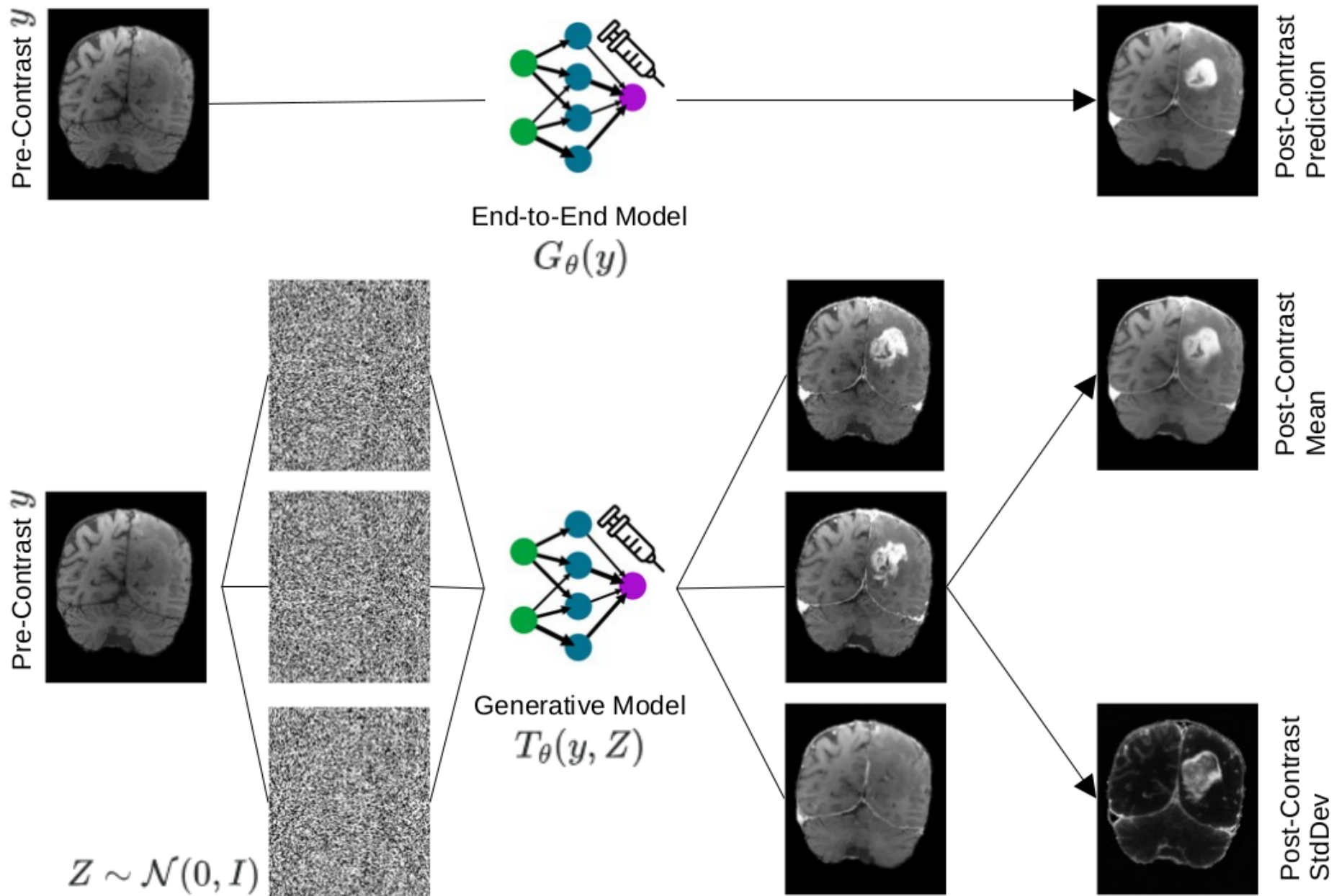
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Bayesian OT Flow Matching: Conditional Wasserstein Distances



3. Bayesian Inverse Problems

1. Bayesian OT Flow Matching

2. PnP Flow

Refs: S. Martin, A. Gagneux, P. Hagemann, G. Steidl,
PnP-Flow: Plug-and-play image restoration with flow matching,
ICLR 2025

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2. PnP Forward-Backward Splitting

Variational model: find minimizer of

$$\mathcal{J}(x) = \underbrace{\mathcal{D}(F(x), y)}_{g(x)} + \underbrace{\lambda \mathcal{R}(x)}_{h(x)}, \quad \lambda > 0$$

◆ h convex, lsc, $g : \mathbb{R}^d \rightarrow \mathbb{R}$ convex, diff. and $\text{Lip} \nabla g = L < \infty$

Fermat's rule:

$$\begin{aligned} 0 &\in \nabla g(\hat{x}) + \partial h(\hat{x}) \\ \hat{x} - \tau \nabla g(\hat{x}) &\in \hat{x} + \tau \partial h(\hat{x}) = (I + \tau \partial h)(\hat{x}) \\ \hat{x} &= (I + \tau \partial h)^{-1} (\hat{x} - \tau \nabla g(\hat{x})) \\ &= \text{prox}_{\tau h} (\hat{x} - \tau \nabla g(\hat{x})) \end{aligned}$$

Forward-Backward Splitting (FBS): (also called proximal gradient algorithm)

$$x^{(r+1)} = \text{prox}_{\tau h} (x^{(r)} - \tau \nabla g(x^{(r)}))$$

Convergence for $\tau \in (0, \frac{2}{L})$:

$$\mathcal{J}(x^{(r)}) - \mathcal{J}(\hat{x}) = \mathcal{O}(1/r).$$

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2. PnP Forward-Backward Splitting

From FBS to Plug-and-Play FBS:

Algorithm 1 FBS

Init: $x_0 \in \mathbb{R}^d$, $\gamma > 0$

Iterations: For $n = 0, 1, \dots$

$$z_n = x_n - \gamma \nabla (\mathcal{D} \circ F)(x_n)$$

$$x_{n+1} = \text{prox}_{\gamma \mathcal{R}}(z_{n+1})$$

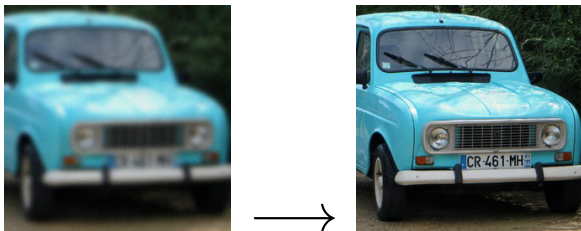
Algorithm 2 PnP-FBS

Init: $x_0 \in \mathbb{R}^d$, $\gamma > 0$

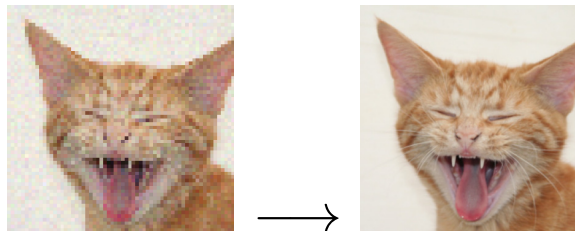
Iterations: For $n = 0, 1, \dots$

$$z_n = x_n - \gamma \nabla (\mathcal{D} \circ F)(x_n)$$

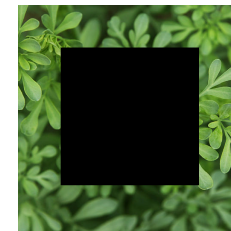
$$x_{n+1} = \textcolor{red}{D}(z_n)$$



Deblurring



Superresolution



Inpainting

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Algorithm 3 PnP-Flow Matching

Input: Pre-trained network v^θ , $T_t^\theta = \text{Id} + (1 - t)v_t^\theta$, $(t_n)_n$ with $t_n = n/N$ and adaptive stepsizes $(\gamma_n)_{1 \leq n \leq N}$.

Initialize: $x_0 \in \mathbb{R}^d$.

Iteration For $n = 0, 1, \dots, N$

$$z_n = x_n - \gamma_n \nabla (\mathcal{D} \circ F)(x_n)$$

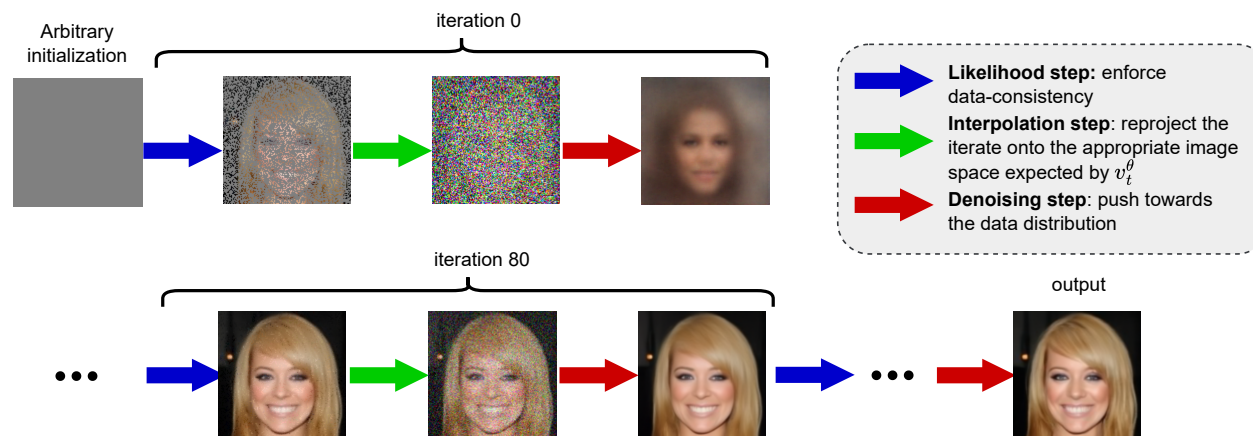
$$\tilde{z}_n = (1 - t_n)\varepsilon + t_n z_n, \varepsilon \sim \mu_0$$

$$x_{n+1} = T_{t_n}^\theta(\tilde{z}_n)$$

Gradient step

Interpolation step

PnP denoising step



Theorem. Assume that the vector field v^θ is continuous. Let the time sequence $(t_n)_{n \in \mathbb{N}}$ satisfy $\sum_{n=0}^{\infty} (1 - t_n) < +\infty$ and let $\gamma_n := 1 - t_n$, $n \in \mathbb{N}$. If the sequence $(x_n)_{n \in \mathbb{N}}$ obtained by the algorithm is bounded, then it converges.

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Figure 11: Comparison of restoration methods on AFHQ-Cat: denoising (1st row), Gaussian de-blurring (2nd row), super-resolution (3rd row), free-form inpainting (4th row).

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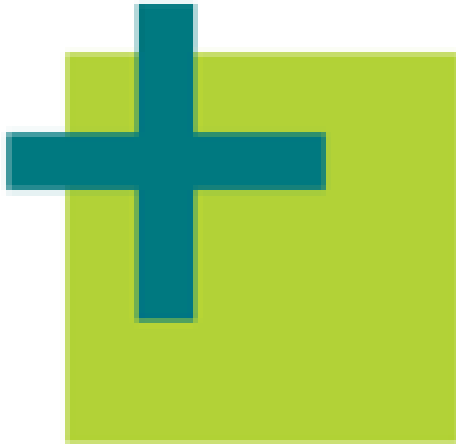
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Berlin Mathematics Research Center

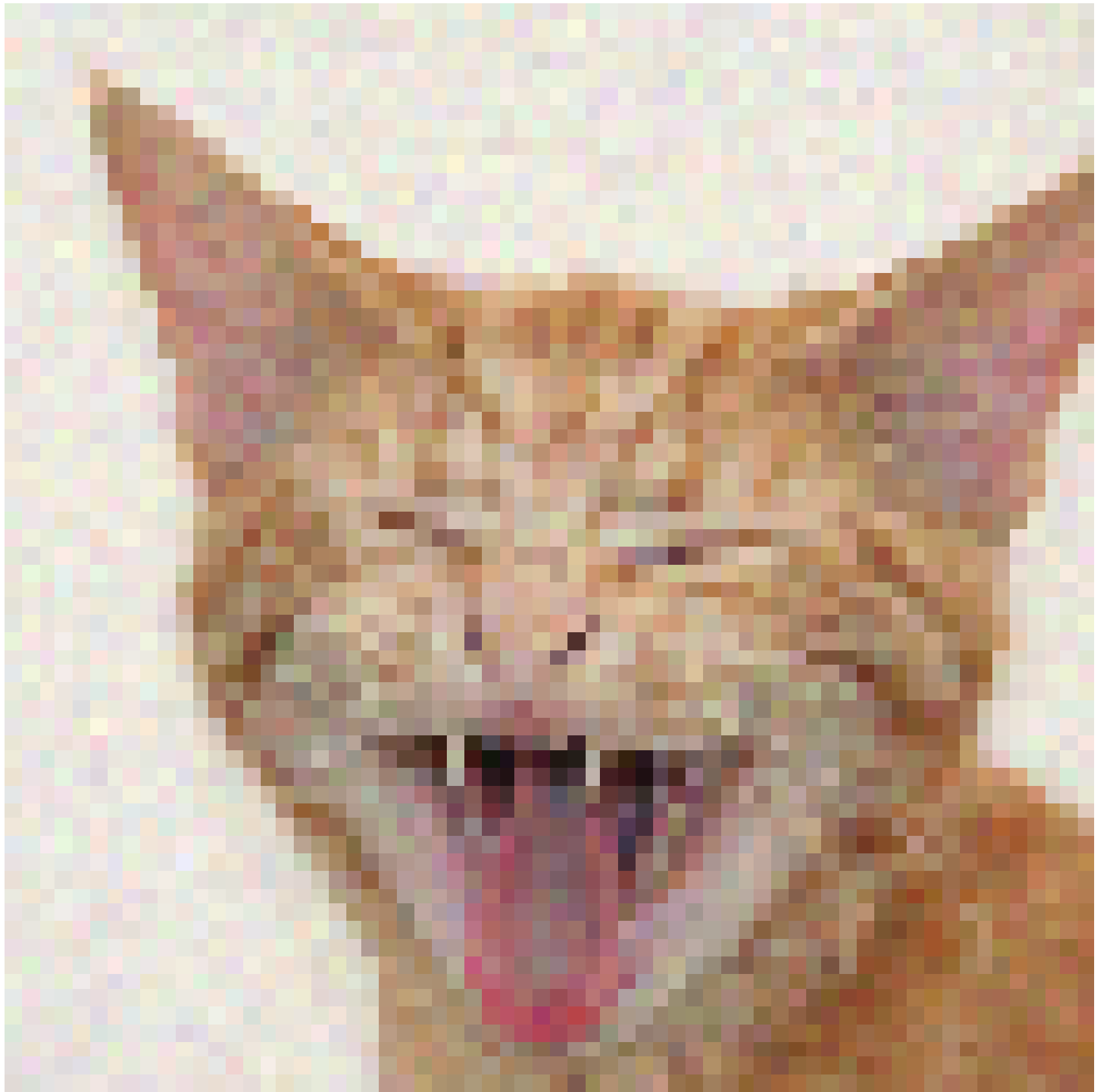
MATH 

Funded under Germany's Excellence Strategy by
DFG Deutsche
Forschungsgemeinschaft







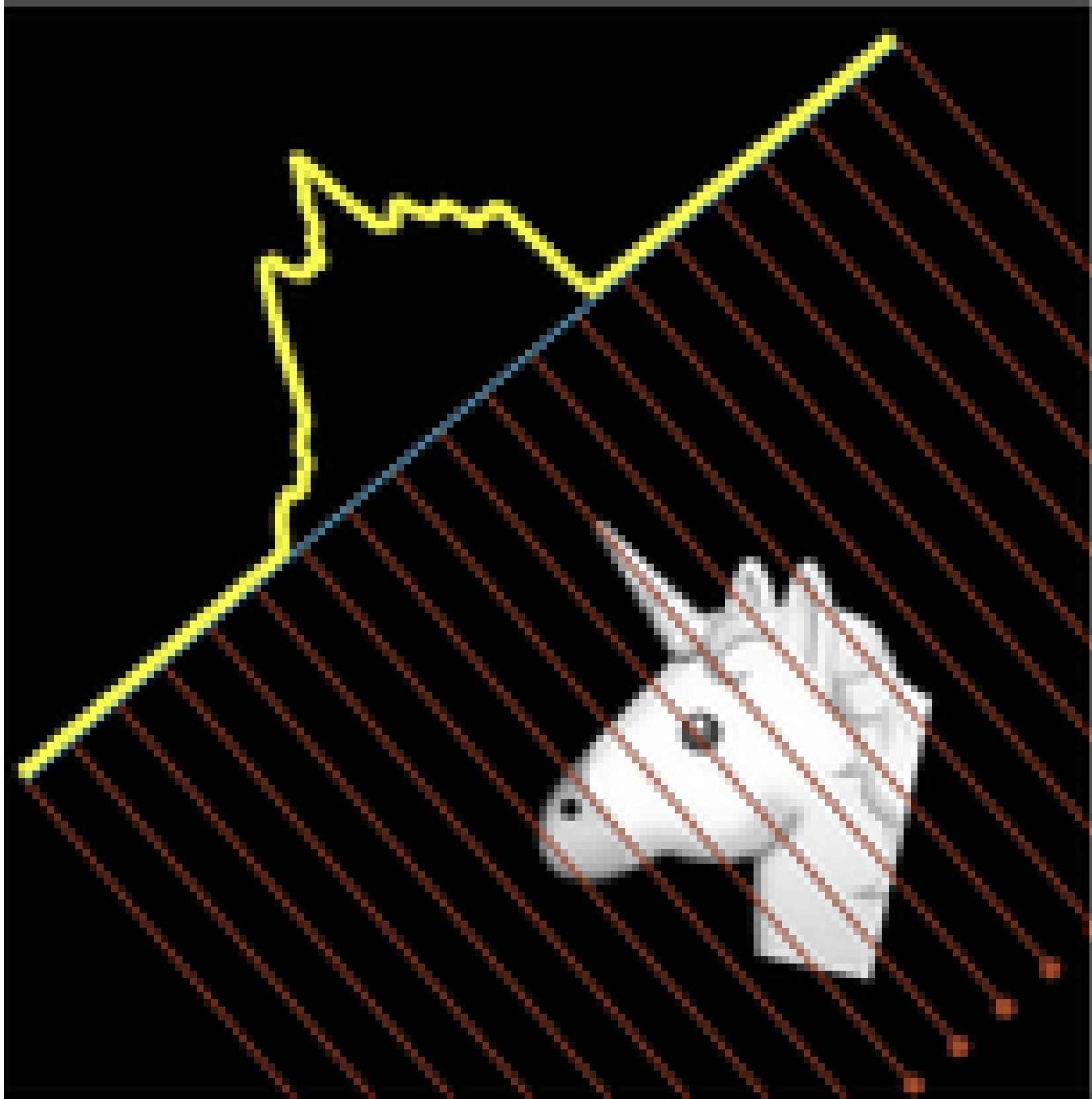


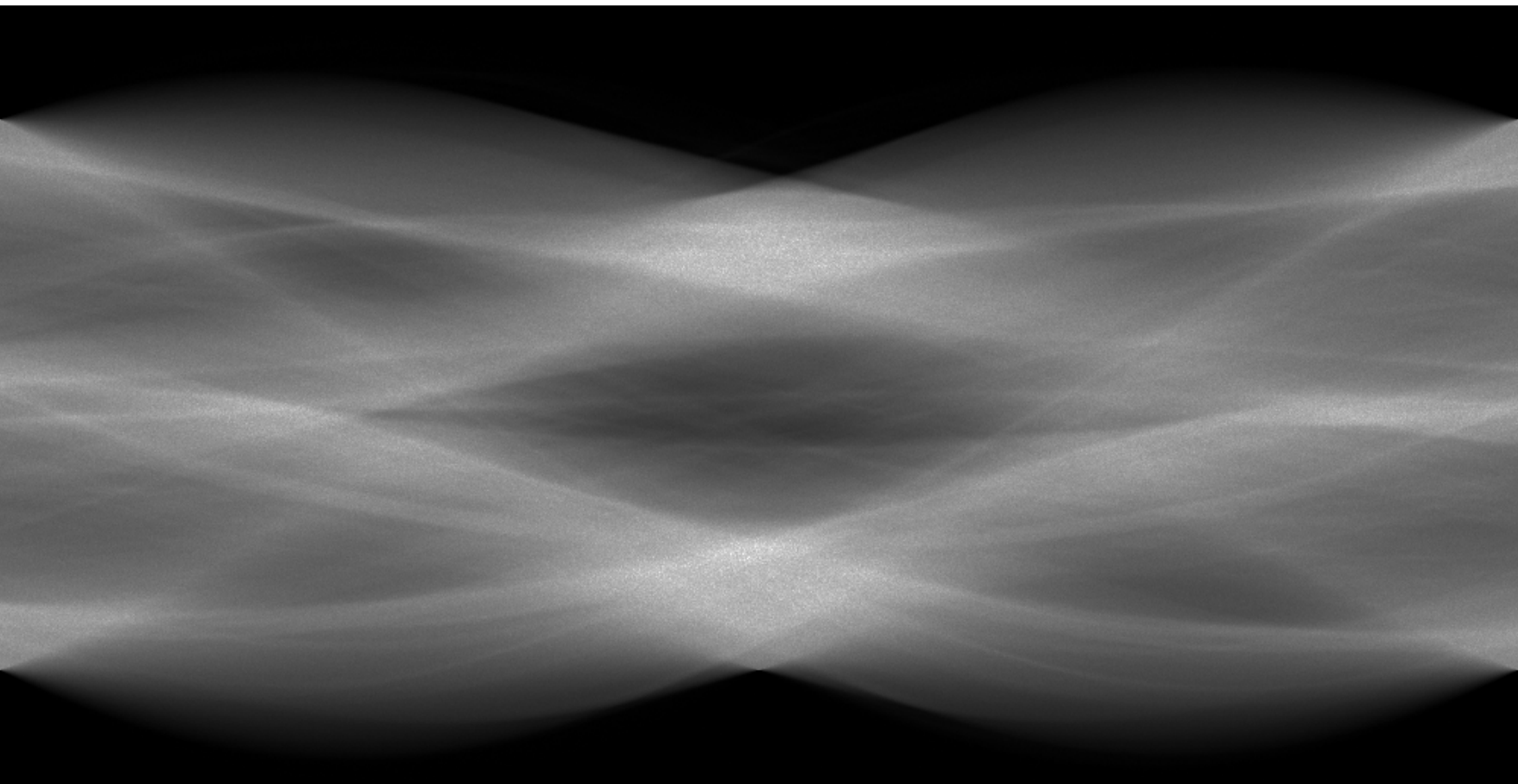


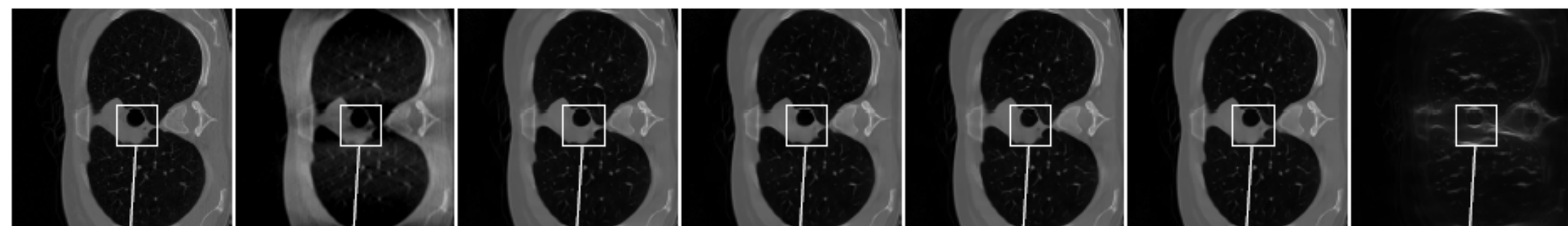












GT

FBP

Reco 1

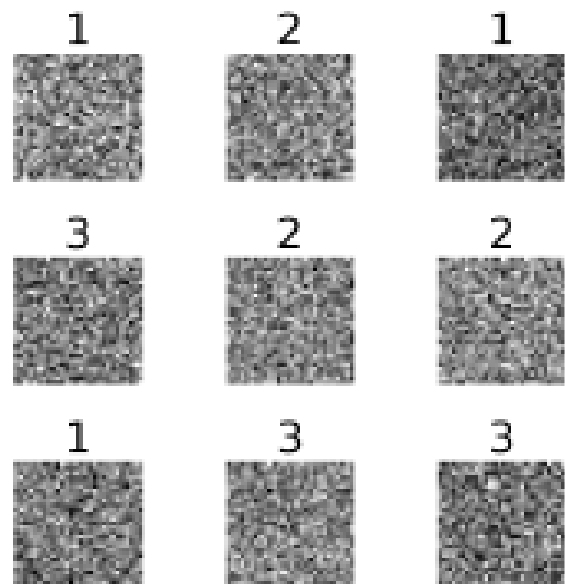
Reco 2

Reco 3

Mean

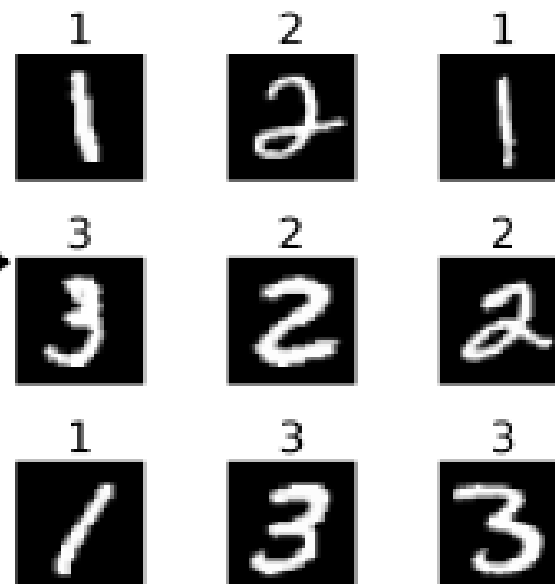
Std

$P_{Y,Z}$



g

$P_{Y,X}$

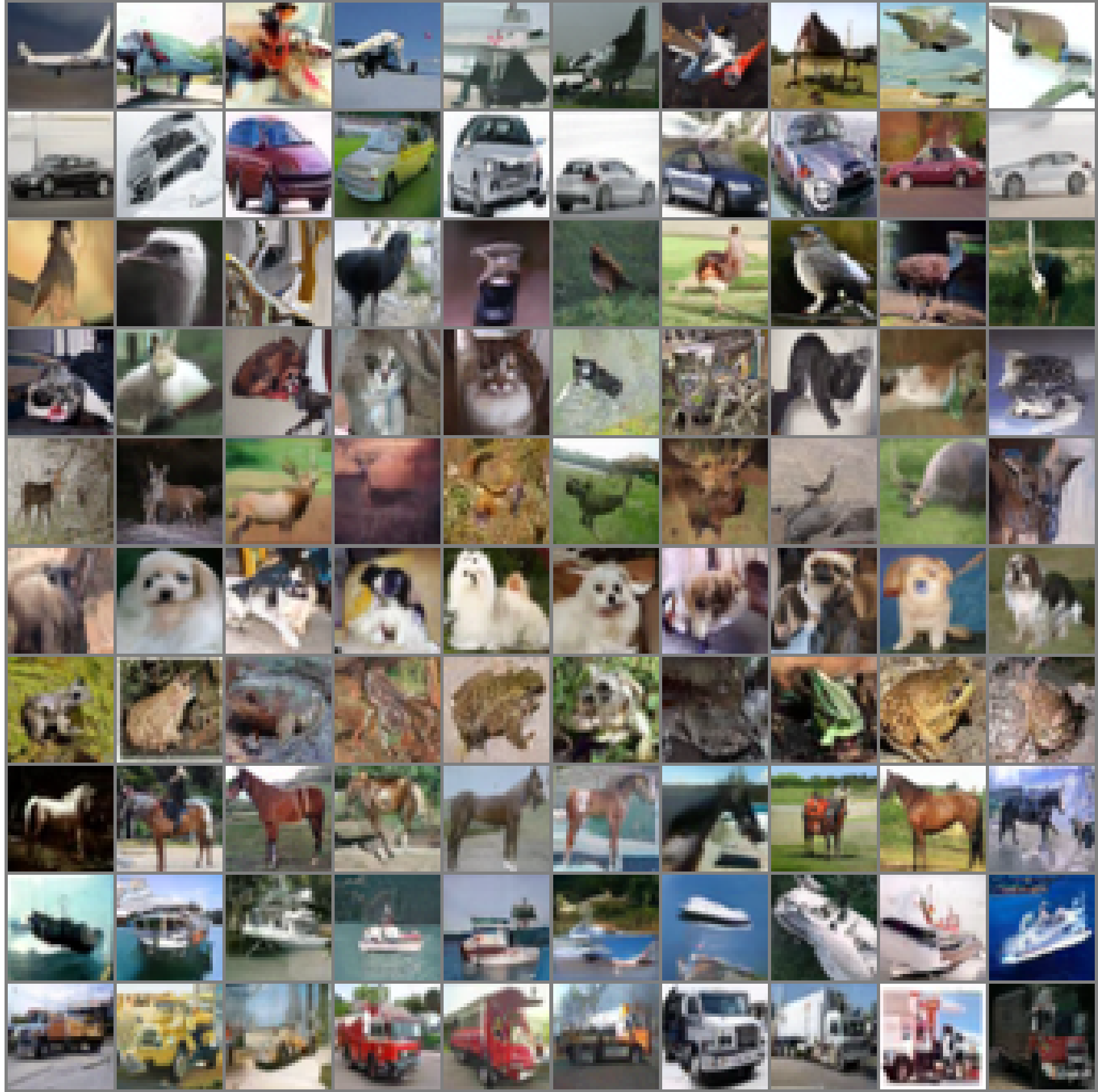


$P_{X|Y=2}$

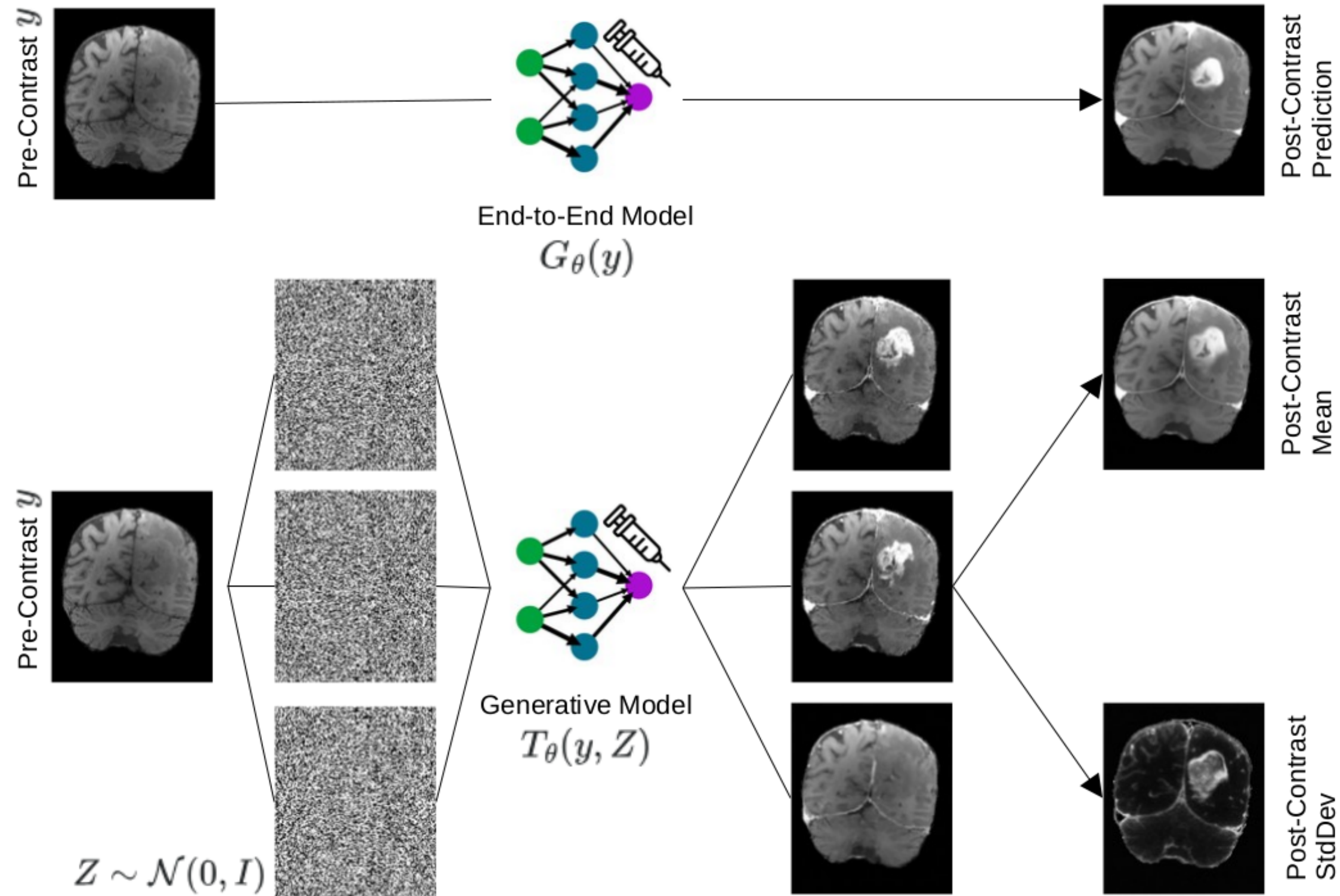






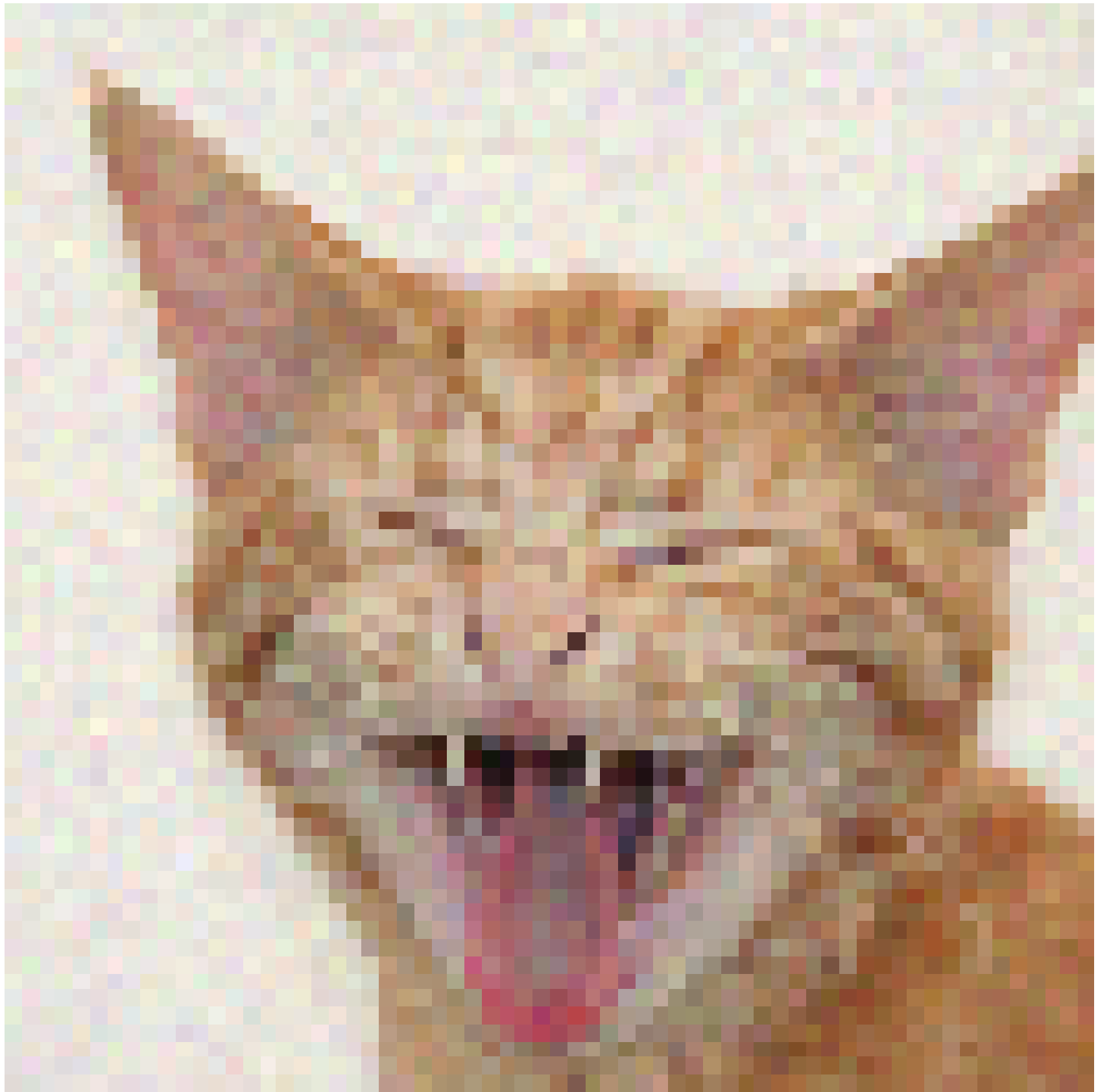
















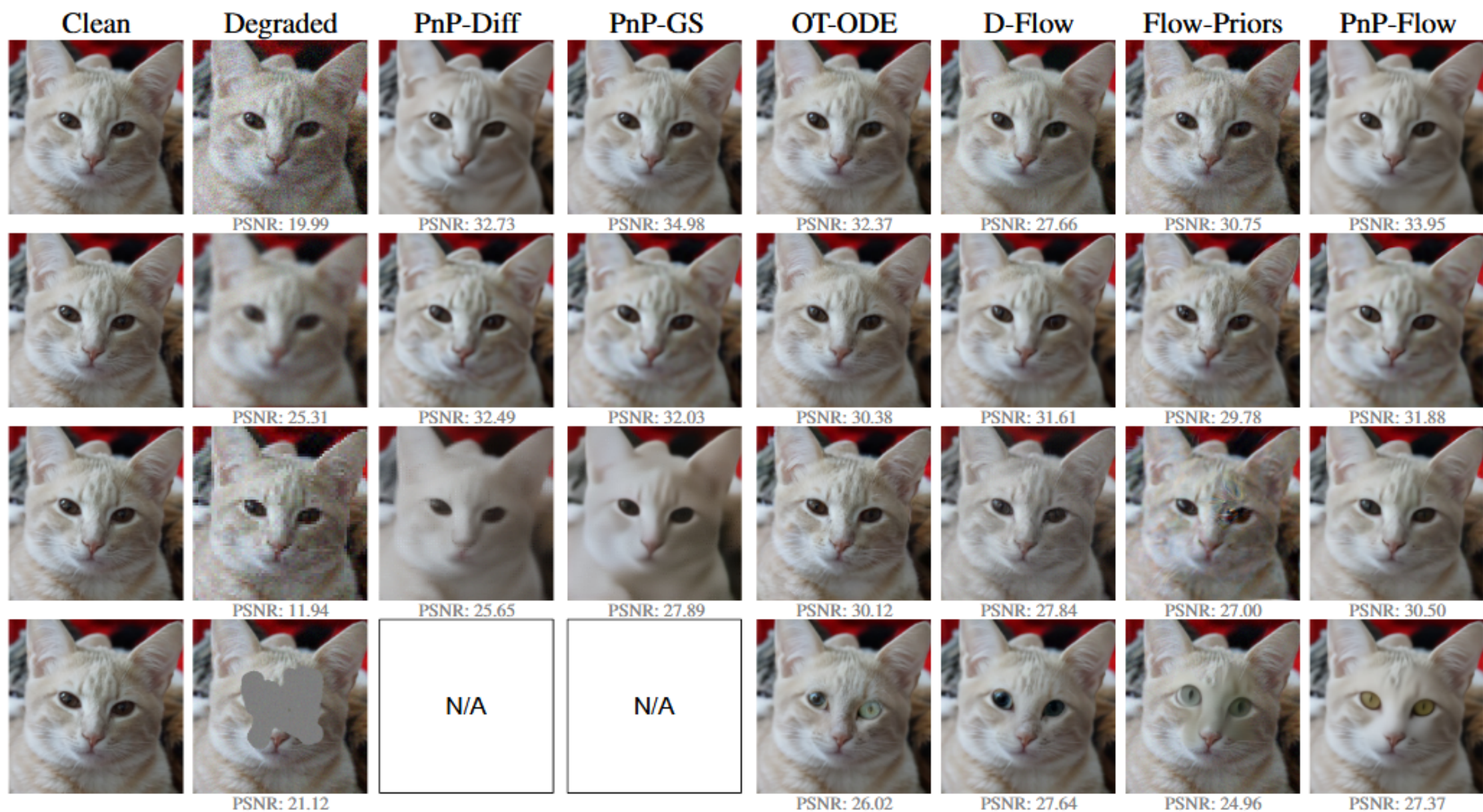


Figure 11: Comparison of restoration methods on AFHQ-Cat: denoising (1st row), Gaussian deblurring (2nd row), super-resolution (3rd row), free-form inpainting (4th row).